

CHRISTIAN ALEMÁN PERICÓN

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WORK EXPERIENCE

New York University Abu Dhabi Research Associate, Economics Division	Abu Dhabi, UAE 2023 – Present
- Developed heterogeneous agent macroeconomic models to inform fiscal and monetary policy using MATLAB and Python. - Led data intensive projects integrating causal inference and forecasting techniques for real time decision making. - Designed and implemented a rigorous evaluation framework to assess the ex-ante/ex-post economic impact of the 2024 French payroll subsidy reform, using a structural DGE macro-model calibrated on confidential administrative microdata. - Designed optimal tax schemes using government data to regulate the purchase of high-cost health treatments in Denmark. - Applied machine learning methods (Least Squares Learning, Self Fulfilling Equilibria) and deep learning architecture to large-scale longitudinal data sets, developing benchmarks to assess forecasts of aggregate labor market expectations and health outcomes. - Applied deep learning architectures (CNNs, GANs, Autoencoders) to solve structural decision models, analyzing model behavior and failure modes to guide research priorities. - Delivered 15+ academic talks and policy presentations across 10+ countries showcasing the above listed projects. - Authored research papers on causal inference, quantitative policy evaluation, and AI-driven decision systems for top-tier peer-reviewed journals (AEJ Macroeconomics, Econometrica, Review of Economic Studies), contributing to the development of novel evaluation methodologies.	
Université Libre de Bruxelles (ULB) – ECARES Research Fellow, Economics Department	Brussels, Belgium 2022
- Applied econometric methods (Generalized Method of Moments, Monte Carlo simulations, Non-parametric Estimation) using MATLAB to study income and wealth inequality resulting from inter-generational transfers, bequests and migration choices. - Developed a DSGE macro-model to study the effects of housing market shocks on household consumption and savings.	
Barcelona School of Economics Predoctoral Researcher	Barcelona, Spain 2018 – 2022
- Co-developed and published a novel econometric evaluation methodology (Stage-Based Identification) to estimate real-time effects of nationwide policies across health, environment, and labor domains; routines publicly available in R, MATLAB, and Python. - Constructed a multi-phased dynamic macro-model in MATLAB to estimate aggregate effects of HIV in developing economies. - Developed a framework to identify match-quality in labor markets using revealed preferences, improving match predictions by 40%. - Led an Advanced Research Methods course for 40 students, focusing on data analysis using Python and R. - Managed projects for government, think tank and corporates, aligning research outputs with stakeholder's strategic priorities.	
Bank Sabadell Research Assistant, Department of Quantitative Analysis	Barcelona, Spain 2018
- Developed and benchmarked time series models (VAR, ARCH, GARCH) to forecast key macroeconomic and financial variables (inflation, GDP, credit) under risk scenarios, producing structured performance assessments for internal strategy teams. - Estimated a Vector Error Correction model to monitor and forecast credit card consumption trends in Mexico. - Evaluated macro-prudential and monetary policy using agent based models and leading indicators for industrial output and GDP. - Collaborated with risk management, strategy and business intelligence teams to align economic outlooks with business goals.	
Halle Institute for Economic Research Research Intern, Department of Macroeconomics	Halle (Saale), Germany 2017
Developed a DSGE input-output model to identify the determinants of long-term growth processes in Germany.	

EDUCATION

Universitat Autònoma de Barcelona, University of Pennsylvania Ph.D. Economics (Cum Laude)	Spain, USA 2018 – 2022
Université Paris I Panthéon-Sorbonne M.Sc. Master in Models and Methods of Quantitative Economics	France 2016 – 2018

SCHOLARSHIPS / GRANTS

Ludwig Maximilian University of Munich and NYU Collaboration Grant	2025
New York University Postdoctoral Research Fund Award	2023 – 2026
FPI PhD Scholarship from the Spanish Ministry of Science	2018 – 2022
Erasmus Mundus Scholarship: Full funding to pursue Master's Degree	2016 – 2018

SKILLS / ADDITIONAL INFORMATION

Programming & Tools:	Python R, Julia, Eviews, SQL, HPC, AWS, Bloomberg, ML/AI: PyTorch/TensorFlow familiarity, LLM evaluation
Linguistic Proficiency:	Spanish (Native), English/French/German (Fluent), Italian/Catalan (Intermediate)
Citizenship:	Spain, UAE Golden Visa holder

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PUBLICATIONS AND RESEARCH ABSTRACTS

The following abstracts summarize key projects I published

Stage-Based Identification of Policy Effects (2026) joint with Christopher Busch, Alexander Ludwig and Raúl Santaaulàlia-Llopis

We develop a new econometric method that identifies the effects of public policy implemented nationwide---i.e. across all regions at the same time. Starting point is the insight that the dynamics of many outcome variables can be tracked over stages. A stage is defined as the location of a regional outcome on a reference outcome path. Our method consists of a normalization that maps the time-path of regional outcomes onto a reference outcome path using only pre-policy data. After normalization, the pre-policy outcome paths mapped onto the reference path are identical across regions which implies that the normalization controls for pre-policy regional heterogeneity (the so-called "parallel trends") without taking a stand on its source, (un)observability or (non)constancy. Since regions can differ by stage at any point in time, the normalization uncovers variation in the stage at the time of policy implementation. We use this stage variation at the time of policy implementation to identify the policy effects. We validate our method with a set of Monte-Carlo experiments. We show several applications: public health stay-home policies in Spain (lockdown), the effects of approval of oral contraceptives in 1960 in the U.S. on women's fertility and career choice and the effects of growth policy (e.g. German Reunification).

On the Macroeconomic Neutrality of Employer and Employees Social Contributions (2026) Etienne Wasmer, Pranav Mimani and Raúl Santaaulàlia-Llopis

This paper revisits the belief that the mix of employer and employee social security contributions is neutral for employment and equilibrium wages. We establish the conditions required for neutrality across competitive settings, institutional minimum wages, progressive taxation, and search-and-matching frictions. We show that statutory invariance breaks down due to a single structural tax asymmetry wedge operating almost identically across these frameworks. Consequently, tax neutrality is restricted to a strict knife-edge case that is comprehensively violated in modern labor markets, rendering statutory non-neutrality highly relevant quantitatively. Our findings offer critical implications for designing contemporary welfare and workfare policies.

Employment and Human Capital Investment Responses to Payroll Tax Reduction (2025) joint with Etienne Wasmer and Pranav Mimani

To counter the negative effects on employment for low-skilled workers, some countries have developed policies that reduce social contributions for low wage workers. We examine the effects of such policies on employment, the incentives to invest in employee training, and salary progression. We propose a model featuring dual heterogeneity of workers and firms, as outlined by Shimer and Smith (2000), to which we add specific human capital accumulation. We calibrate the model on France, where the policy of tax relief, initiated in 1993, has reached a record of 80 billion euros (2.5% of GDP). We find that the reductions in social contributions for low wage workers disincentivize human capital investments and contribute significantly to the high concentration of workers who prefer to keep earning the minimum wage. Our simulations suggest that a one percent reduction of the social contribution subsidy increase average wages by 4% and net employment by 2.5%.

Double Selection in Causal Inference (2025) joint with Christopher Busch, Raúl Santaaulàlia-Llopis and Tanya Surovtseva

A long-standing question in research using natural experiments for causal inference is the appropriate functional form of the outcome variable, often involving a choice between level or logarithmic formulations. This choice is crucial because the assumption of a single constant selection bias (CSB)—or, equivalently, a parallel time trend—for identifying the policy effect cannot hold simultaneously in both formulations. To address this issue, we propose a novel double-selection (DS) econometric approach, in which units face two types of selection: a level shifter (single CSB in levels) and a proportional shifter (single CSB in logs). We show unique identification of the DS and, consequently, the causal effects. We propose a DS estimator in levels and conduct inference.

Evaluating the Effectiveness of Policies Against a Pandemic (2024) joint with Christopher Busch, Alexander Ludwig and Raúl Santaaulàlia-Llopis

We apply a novel method to evaluate policies against a pandemic. In the context of the application, the essence of our approach is the insight that epidemic dynamics are best tracked over stages, rather than over time. The application focuses on the nationwide stay-home policy enacted in Spain against the Covid-19 pandemic. We find that the policy saved 16.8% of lives relative to the number of deaths that would have occurred had it not been for the policy intervention. Its effectiveness evolves with the epidemic and is larger when implemented at earlier stages.

Kramer vs. Kramer. On the importance of Children and Divorce Filings for Understanding Divorce Rates in the USA (2024)

I document that approximately 70% of divorce filings in the 1970's were done by wives in the United States. Since then, this figure has experienced a large decline, reaching 56% in 2015. At the same time, divorce rates sharply increased from 1960 until the mid 1980's and have declined since then. I construct a life cycle model of endogenous marriage and unilateral divorce with endogenous labor supply and savings that jointly explains these facts. I use my model to measure the contribution of changes in the gender-wage gap, property division laws and child custody arrangements in explaining the divorce patterns over time. First, the reduction in the gender-wage gap generates two opposing effects. On the one hand, the reduction of the gender-wage gap increases the value of divorce for married women and, on the other hand, unmarried women become more selective in the marriage market thus raising the quality of newly formed matches. Second, children increase the value of divorce for the custodial parent; so a higher probability of getting child custody raises the chances of filing for divorce. Third, a higher share of assets assigned to wives upon divorce can either increase or decrease divorce rates by altering the savings decision of the household. My model accounts for approximately 50 per cent of the decline in divorce filings and 70 per cent of the variation in divorce rates between 1970 and 2015. I find that the decrease in the gender-wage gap and the increase in the probability of getting child custody for men are major drivers behind the changes in divorce rates and in divorce filings, respectively. Importantly, I find that failure to match who files for divorce can lead to opposite counterfactual results.

A Quantitative Theory of the HIV Epidemic: Education, Risky Sex and Asymmetric Learning (2024) joint work with Daniela Iorio and Raúl Santaaulàlia-Llopis

Using micro survey data, we show that the relationship between education and the probability of HIV infection is U-shaped (positive-zero-positive) over the course of the epidemic. In contrast, the relationship between education and knowledge about the process of HIV infection follows an inverted U-shaped pattern. We develop a non-stationary quantitative macroeconomic theory with heterogeneous agents that is consistent with these facts. Our theory endogenizes the entire course of the HIV epidemic across its different stages: a pre-HIV epidemic stage; a myopic HIV stage in which agents are not aware of the process of HIV infection; a learning stage in which agents heterogeneously---across education groups---learn about the process of infection; and an anti-retroviral (ARV) stage that modifies the effects of HIV infection on individuals. We show that asymmetric learning is key to reproduce both the micro patterns that we document and the aggregate evolution of the HIV epidemic. In further counterfactual experiments, we assess the effects of an early understanding of the virus and its mode of infection, improvements in the composition of education, the earlier (and universal) adoption of ARVs and the use of PrEP to prevent further spread.